

MathRules

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simple math rules I always mess up on. Sometimes I describe the practical application of this math, I encourage you to glance over it to see why you need to learn math.

1 Representation of Numbers

The following math expressions keep popping up during my time at Berkeley

- $\sqrt{x} = x^{1/2}$
- $-x = \frac{1}{x}$
- It follows that $\sqrt{-x} = \frac{1}{x^{1/2}}$
- Taylor Series
- Odd:= $2k + 1$, Even:= $2k$

If (convert positive constant into square root):

$$c \rightarrow c^2 \rightarrow \sqrt{c^2}$$

$$5 = 25 = \sqrt{25}$$

Elif (convert negative constant into square root):

$$-c \rightarrow -(c^2) \rightarrow -\sqrt{c^2}$$

$$-5 = -(25) = -\sqrt{25}$$

2 Division

- Dividing a positive number a by negative $-b$ will result in $-\frac{a}{b}$, likewise for multiplication.
- $\frac{c}{0}$ is not undefined, because $a|0 = 0 \cdot c = a \implies -T$

Simplifying the radical and working in fractions:

$$\begin{aligned}\frac{3}{\sqrt{10}} \cdot -\frac{1}{\sqrt{20}} &= -\frac{3 \cdot 1}{\sqrt{10 \cdot 20}} = -\frac{3}{\sqrt{200}} = -\frac{3}{\sqrt{100 \times 2}} = -\frac{3}{10\sqrt{2}} = \\ &= -\frac{3}{10\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = -\frac{3\sqrt{2}}{10 \times 2} = -\frac{3\sqrt{2}}{20}\end{aligned}$$

3 Difference of Squares

Say we have

$$\frac{2}{-2 - 4i} \cdot \frac{-2 + 4i}{-2 + 4i}$$

We are rationalizing the lhs. Notice the identity $(a + b) \cdot (a - b)$. This allows us to do:

$$(a^2 - b^2)$$

In this case:

$$(4 - 16i)) = 4 - 16(-1) = 4 + 16 = 20$$

Thus:

$$\frac{-4 + 8i}{20}$$

4 Factoring

Quadratic factoring

AC Method Algorithm

$$ax^2 + bx + c$$

- Split bx into two terms, i.e. into coefficients that satisfy the conditions:

– coefficients add up to bx

– coefficients multiply to c

E.g., Let b_1, b_2 be any real number that satisfies the conditions of $b_1 + b_2 = bx \wedge b_1 \cdot b_2 = ac$

$$\begin{aligned}3x^2 + 22x + 24 &= \\ 3x^2 + 4x + 18x + 24 &= \\ (3x^2 + 4x) + (18x + 24) &= \\ x(3x + 4) + 6(3x + 4) &= \\ (x + 6)(3x + 4) &= \end{aligned}$$

How does $r^2 - 1$ become $r = \pm 1$?

$$r^2 - 1 \rightarrow r^2 = 1 \rightarrow \sqrt{r^2} = 1 \rightarrow r = 1$$

More formally, simplify squared roots with $\sqrt{\quad}$ on both sides

5 Laws of Exponents

Website Quick Cheat sheet

6 Square Roots

- Square roots only distribute over multiplication and division
- Properties of Square roots

Product Property: $\sqrt{a} \times \sqrt{b} = \sqrt{a \times b}$

Dealing with addition or subtraction: $\sqrt{9} + \sqrt{16} \rightarrow 3 + 4 = 7$ If the roots are the same, add the coefficients in front:

$$\begin{aligned}\sqrt{18} + \sqrt{50} &= \\ 3\sqrt{2} + 5\sqrt{2} &= \\ 8\sqrt{2} &\end{aligned}$$

More formally, simplifying square roots algorithm:

- Find the largest square root within the number a , call it b . Where b has the condition $a = b \times c$ with b as a perfect square of the form d^2
- Then set it up as $\sqrt{b \times c}$
- $\sqrt{b} \times \sqrt{c} = d\sqrt{c}$

Example:

$$\sqrt{28} = \sqrt{4 \times 7} = \sqrt{4} \times \sqrt{7} = 2\sqrt{7}$$

Dividing a square root Property

$$\sqrt{2} \div 2 = \sqrt{\frac{2}{4}} = \sqrt{\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

More formally:

$$\text{Let } a, b \in R : \sqrt{a} \div b = \frac{\sqrt{a}}{\sqrt{b^2}} = \sqrt{\frac{a}{b^2}} = \text{new representation}$$

$$\text{Applied : } \frac{\sqrt{32}}{2} = \sqrt{\frac{32}{2^2}} = \sqrt{8} = 2\sqrt{2}$$

Definition: The representation of square roots across all numbers

For positive numbers, square roots yield two solutions:

- a positive and a negative root
- while cube roots provide only one solution (which retains the sign of the original number)

Notably, negative numbers under even roots result in imaginary numbers, while odd roots can yield real negative results. Source

1. Simplifying Radicals

$$\sqrt{27/2} = \sqrt{\frac{9 \cdot 3}{2}} = \frac{3 \cdot \sqrt{3}}{\sqrt{2}} = \frac{3 \cdot \sqrt{3}}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{3\sqrt{6}}{2}$$

2. Further simplification example

$$3/2 \cdot \frac{2}{3\sqrt{6}} = \frac{3}{2(3\sqrt{6})} \cdot \frac{2}{3\sqrt{6}} = \frac{6}{2(3\sqrt{6})} = \frac{6}{6\sqrt{6}} = \frac{1}{\sqrt{6}}$$

3. Wild Interaction of diving a constant by square root, then adding to a constant:

$$\sqrt{3} \div 3 = \frac{\sqrt{3}}{3} = \frac{\sqrt{3}}{3} + 2 = \frac{\sqrt{3}}{3} + \frac{2}{1} = \frac{\sqrt{3}}{3} + \frac{6}{3} = \frac{6\sqrt{3}}{3}$$

7 PEMDAS/BODMAS

Whats the order of operations for $c \cdot k^n$?

- Apply k^n to find a
- Then proceed with $c \times a = z$

8 Complex Numbers

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$$

Steps:

$$ac + adi + bic + bdi^2 = ac + adi + bdi + bd(-1)$$

$$(ac - bd)(ad + bd)i$$

Notes:

- $bidi, bd(i^2)$ are the same identity.
- we use $i^2 = -1$ identity

- $\sqrt{-4} \rightarrow \sqrt{(4 \cdot -1)} \rightarrow \sqrt{4} \cdot \sqrt{(-1)} = 2 \cdot i = 2i$. **WHY?**
 - Rule applied: $\sqrt{(-n)} = \sqrt{n} \cdot i$
 - We represent $\sqrt{-1} = i$ because a square root can never be negative, and thus that allows us to represent the negative as i
- A constant divided by i , is equivalent to saying $-i$ times that constant:

$$\frac{c}{i} = -ic$$

$$\frac{c}{i} \cdot \frac{i}{i} = \frac{ic}{i^2} = \frac{ic}{-1} = -1(ic) = -ic$$

Moral of the story: Break it up when you can to simplify expressions easier.

$$\frac{4\sqrt{-4}}{2} \rightarrow \frac{4}{2} \pm \frac{2i}{2} \rightarrow 2 \pm i$$

8.1 Back Substitution with Imaginary Numbers

Say we have a system where $x_1 = \frac{-2+i}{-i}$. The following intuition should be built:

$$\begin{aligned} \frac{-2+i}{-i} \text{ get rid of } -i \rightarrow \frac{-2+i}{-i} \cdot \frac{i}{i} &= \frac{-2i+i^2}{-i^2} = \frac{-2i+(-1)}{-(-1)} = \frac{-2i-1}{1} \\ &= \frac{-1-2i}{1} = -1-2i \end{aligned}$$

9 Synthetic Division

- Guide
- Coding Guide

We are given the problem $-\lambda^3 + 12\lambda^2 - 45\lambda + 50$. With synthetic division we do the following:

$$\begin{array}{r|rrrr} 1 & -1 & 12 & -45 & 50 \\ & & -1 & 11 & -34 \\ \hline & -1 & 11 & -34 & 16 \end{array}$$

- Quick rule, once you try a candidate from the set of 50's factors, and if it does not go to 0, you move onto the next candidate in the set of factors.
- So in this example, if 1 didnt work, we would try another candidate, lets do 5.

$$\begin{array}{r|rrrr}
 & -1 & 12 & -45 & 50 \\
 5 & & -5 & 35 & -50 \\
 \hline
 & -1 & 7 & -10 & 0 \\
 \hline
 x^3 - 4x^2 - x + 4
 \end{array}$$

$$\bullet \text{ When you choose the root four, you get } \begin{array}{r|rrrr}
 & 1 & -4 & -1 & 4 \\
 4 & & 4 & 0 & -4 \\
 \hline
 & 1 & 0 & -1 & 0
 \end{array}$$

- Leaving you with a remainder of $x^2 - 1$. This is another root. (Recall that we drop the power in synthetic division). This leaves us with ± 1
- It was interesting doing this because I have never encountered that before, were on 4 we get rid of two variables.
- This calculator shows steps

10 Back substitution (Linear Algebra Applications)

Say we have the system:

$$A = \begin{bmatrix} -i & 2 & -1 \\ 0 & -i & 1 \\ 0 & -1 & -i \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -ix_1 + 2x_2 - x_3 \\ -ix_2 + x_3 \\ -x_2 - ix_3 \end{bmatrix}$$

Building this intuition takes time. You want to select the equation that yields an easy way to solve for a variable and by induction we can apply it to solve other systems of equations. In this case, the following equation is worth our time $-ix_2 + x_3$.

$$-ix_2 + x_3 = 0 \rightarrow x_3 = -ix_2$$

We often say that this equation has two unknowns. There is a dependent relationship between the two. Specifically. $-ix_2 = x_3 = x_2 = x_3/i$. Where x_2 is the pivot (independent, and thus we say $x_2 = 1$). We say x_3 is the free variable. From here, it follows:

$$\begin{aligned}
 x_3 = ix_2 = i(1) = i &\implies x_3 \in \text{Row 3} = -x_2 - ix_3 = -(1) - i(i) \\
 &= -1 - i^2 = -1 - -(1) = -1 + 1 = 0 \\
 &\implies R3 \text{ redundant}
 \end{aligned}$$

$$\begin{aligned}
 -ix_1 + 2x_2 - x_3 &= -ix_1 + 2(1) - i = x_1 = \frac{-2 + i}{-i} \\
 x_1 &= -1 - 2i
 \end{aligned}$$

See 8.1 for dealing with $\frac{-2+i}{-i}$

We have just shown back substitution, but why were we solving this? Whats the meaning?

Algo:

1. Find eigenvalues
2. Find eigenvector
3. Write in general solution with found eigenvalues $e^{(roots)} \cdot E_n$
4. Set the eigenvalues found in step 3 equal to $\mathbf{x}(0)$, then reduce the row in the upper triangular form, and back substitute. This is your answer.

The following algorithm is useful because:

- The algorithm is literally what LAPACK (the library under NumPy, Octave, MATLAB) executes. You are learning the source code conceptually.
- Sometimes we need to find A^k . A^k is seen in Markov chains, graph walks, ML, and this yields $O(kn^3)$ runtime from $A^k = A + A^2 + \dots + A^k$
- Eigendecomposition and diagonalization allows us to do $A = PDP^{-1}$, meaning when we need A^k , $A^k = PD^kP^{-1}$ will yield $O(n)$ time because only the diagonal entries will be raised to k

11 Math 54 Theorems

11.1 Chapter 5.3 - Diagonalization

Definition

Similarity is defined as:

$$A \sim B \iff \exists P (\det(P) \neq 0 \wedge A = PBP^{-1})$$

By similarity, then $Tr(A) = Tr(B)$

$$Tr(A) = Tr(B)$$

$$Tr(PBP^{-1}) = Tr(PBP^{-1})$$

$$Tr(PP^{-1}B) = Tr(PP^{-1}B)$$

$$Tr(IB) = Tr(IB)$$

In other words, $Tr(A) = Tr(IB)$

Lemma

Transforming the order of multiplication in Traces. We are given the condition that $(XYZ) = ZXY \vee YXZ \vee YZX$ for traces.

11.2 Chapter 5.5 - Eigenvalues & Eigenvectors

Lemma

We use the quadratic formula:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Concepts:

- The eigenspace is a **subspace**, so we just need a basis vector. Any nonzero scalar multiple works.
- When you have a 2×2 matrix, and you have solved for one eigenvalue (v_1), the conjugate pair $\lambda = a - bi$ gives v_2 , by *conjugate*(v_1)
- When applying quadratic formula to a complex equation, you express λ 's as 1.

11.3 Orthogonality

Definition

An orthogonal set $S := \{v_1, \dots, v_m\} \subset R^n$ s.t. $\forall v_i, v_j \in S, \langle v_i \cdot v_j \rangle = 0, (i \neq j)$

11.4 Projection

Definition

The projection of y onto u is: $proj_y u = (\frac{y \cdot u}{u \cdot u})u$

y orthogonal to u is $y - \hat{y}$

11.5 Imaginary Numbers

Rationalizing the denominator

Definition

$$i = \sqrt{-1} \wedge i^2 = -1$$

11.6 Polynomials and Roots

Ran through this when doing cofactor expansion to find eigenvalues of the sum

Definition

Cofactor expansion is defined as:

$$C_{ij} = (-1)^{i+j} M_{ij}, \text{ where } M_{ij} := \det(\text{Minor Matrix})$$

Minor Matrix is formed by deleting the i-th row and j-th column.

- Source

Let the following equation be the first row and third column. We are in 3×3 dimension. The way to set it up in this case is CM_{ij} to apply the rule of $+$ $-$ $+$ to the CM

$$C \cdot (-1)[(-1)(-1) - (4 - \lambda)(-1)] = 1[1 - (-4 + \lambda)] = 1[1 + 4 - \lambda] = 5 - \lambda$$

$$\text{Where } C = (-1)^{1+3} \implies (1)(-1) = -1(5 - \lambda) = \lambda - 5 = \text{Third Minor}$$

Then lets assume that we found $CM_{ij} + CM_{ij} + CM_{ij}$ for our 3×3 matrix. Then when we sum them we get $-\lambda^3 + 12\lambda^2 - 45\lambda + 50$. We learnt that this is equivalent to the negation, and we prefer to make the roots positive, so we multiply it by -1 to get $\lambda^3 - 12\lambda^2 + 45\lambda - 50$. Remember both are $\equiv$.

It comes from field of axioms, multiplicative closure of equality over a field.

11.7 Finding Eigenvalues and then Eigenvectors

Very rarely will a course ask you to find the eigenvalues of a matrix that has higher dimension than 3×3 . So there is a computational trick you can do to find the characteristic equation. For dimensions above 3×3 , rely on cofactor expansion. The computational methods are shown here for quick reference.

Sarrus Rule: Sarrus Mnemonic

$$A = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \rightarrow \det(A) = aei + bfg + cdh - gec - hfa - idb$$

Demonstrating cofactor expansion with a case example (I aint latexing all dat):

$$\begin{pmatrix} 4 & -1 & -1 \\ -1 & 4 & -1 \\ -1 & -1 & 4 \end{pmatrix} \rightarrow C_{1j} \Rightarrow \begin{vmatrix} 4-\lambda & -1 \\ -1 & 4-\lambda \end{vmatrix} + (-1)^{1+2} \begin{vmatrix} -1 & -1 \\ (-1)(-1) \end{vmatrix} + (-1)^{1+3} \begin{vmatrix} -1 & 4-\lambda \\ (-1)(-1) \end{vmatrix}$$

$$= (4-\lambda)[(4-\lambda)(4-\lambda)-1] - 1[(-1)(4-\lambda) - (-1)(-1)] - 1[(-1)(-1) - (4-\lambda)(-1)]$$

$$= (4-\lambda)[16-4\lambda-4\lambda+\lambda^2-1] - 1[-4+\lambda-1] - 1[1-(4-\lambda)]$$

$$= (4-\lambda)[\lambda^2-8\lambda+15] - 1[\lambda-5] - 1[5-\lambda]$$

$$= (4-\lambda)(\lambda^2-8\lambda+15) - (\lambda-5) - (5-\lambda)$$

$$= (4-\lambda)(\lambda^2-8\lambda+15) - \lambda + 5 - 5 + \lambda$$

$$= (4-\lambda)(\lambda^2-8\lambda+15)$$

$$= 4\lambda^2 - 32\lambda + 60 - \lambda^3 + 8\lambda^2 - 15\lambda$$

$$= -\lambda^3 + 12\lambda^2 - 47\lambda + 60$$

$$(1)(4-\lambda) \quad -\lambda^2 + 12\lambda^2 - 45\lambda + 50$$

Important Procedural Note: You should choose a row and column s.t. when you do cofactor expansion it will yield less compute time. You do that by choosing the row/column with the most 0's.

11.8 Imaginary Numbers

Rationalizing the denominator

Definition

$$i = \sqrt{-1} \wedge i^2 = -1$$

For fractions, say we have $\frac{1}{i}$ and want to i out of the bottom fraction

$$\frac{1 \cdot i}{i \cdot i} \rightarrow \frac{i}{i^2} \rightarrow \frac{i}{-1} = -i$$

Say we have $\frac{x_2}{i}$ due to some operation, then:

$$\frac{x_2}{i} \rightarrow -ix_2$$

because $\frac{1}{i} = -i$

12 Fractions

Say we have $\frac{1/3}{1/\sqrt{3}}$, this is the same as saying $\frac{1}{3} \times \sqrt{3} = \frac{\sqrt{3}}{3}$. The rule is:

$$\frac{a}{b} = \text{multiplication by its reciprocal } \frac{b}{a} = \frac{a}{b} \times b$$

In our example, $a = 1, b = \sqrt{3}$

If we have $\frac{a}{b} \times \frac{a}{b} \rightarrow \frac{a \times a}{b \times b}$. I.e.:

$$\frac{3}{3} \times \frac{2}{3} = \frac{4}{9}$$

To simplify the following:

$$\frac{1/2}{5/4} = \frac{1}{2} \times \frac{4}{5} = \frac{4}{10} = 2/5$$

- In other words, we multiply the fraction $1/2$ by the reciprocal of $5/4$
- Another example per Keewhan

$$\frac{1/2}{5/4} = 2/5$$

- He said just keep the denominator and numerator, i.e. whats inside.
- **Correction:** $\frac{1/2}{5/4} = \frac{1 \times 4}{2 \times 5} = \frac{4}{10} = 2/5$
- Koreans like to say 5 then divides 2, which I like better than the American linguistic 2 over 5
- For some reason I have been blindly writting any constant turned into a fraction, i.e. lets say 14 into a fraction to be in the form of $\frac{1}{14}$. This is incorrect. tThe correct thing to do is to think: **14 parts, where each part is a whole = $\frac{14}{1}$**

Say we have $1 - 5/2$. 1 can be represented as $2/2$. Implying $1 - 5/2 \equiv 2/2 - 5/2 = -3/2$. The general rule that works for all cases when converting fractions is:

$$n = n \times \frac{d}{d} = \frac{n \times d}{d}$$

$$3 = 3 \times \frac{2}{2} = \frac{6}{2}$$

Say we have $\frac{2/3}{2}$. Then order of operations would be:

$$\frac{2/3}{2} = \frac{2}{3} \times \frac{1}{2} = \frac{2}{6} = \frac{1}{3}$$

Why do I need to remind myself this? Because I tried to do reciprocal magic, dont apply that here.

13 Irrationals

Whenever we have $\sqrt{1/3}$, we express it as $1/\sqrt{3}$

14 Decimal Arithmetic

Guide

15 Calculus

Determining if a function is odd or even

Definite Integral

- Definite Integral
- In pauls example, $\int_a^b x^2 + 1 dx \rightarrow x^3 + x|_a^b$, however, we learnt problems with multiplication that we must apply it before taking the anti-derivative procedure for $\int_a^b x^2 \cdot 1 dx$, i.e. for this it becomes $\rightarrow \int_a^b x^2 \cdot 1 = x^3|_a^b$

Definition

Product Rule

$$f(x) \cdot g'(x) + f'(x) \cdot g(x)$$

Chain Rule in action

$$\frac{d}{dx} [f(x)^{u(x)}] = f(x) \cdot u'(x)$$

$$\begin{aligned} \frac{d}{dx} [e^{x \ln 2}] &= e^{x \ln 2} \cdot \frac{d}{dx} [x \ln 2] = e^{x \ln 2} \cdot \ln 2 \cdot \frac{d}{dx} [x] \\ &= e^{x \ln 2} \cdot (\ln 2)(1) = e^{x \ln 2} \cdot \ln 2 \end{aligned}$$

$$\text{Rewrite/simplify: } \ln 2 \cdot e^{x \ln 2} = \ln(2) \cdot 2^x$$

Applying constants in the middle of differentiating example

$$\begin{aligned} (-2e^{-\theta} \sin(\theta))' &= -2 \times [e^{-\theta} \times \cos(\theta) - e^{-\theta} \times \sin(\theta)] = \\ &= -2e^{-\theta} \cos(\theta) + 2e^{-\theta} \sin(\theta) = 2e^{-\theta} (-\cos(\theta) + \sin(\theta)) \end{aligned}$$

Originally, you only applied the 2 once, when we have two functions and the constant applied, we must distribute it within. I.e. the wrong thinking is: $-2e^{-\theta} \cos(\theta) + e^{-\theta} \sin(\theta)$

Definition

Three Product Rule

If $y = f(x)g(x)h(x)$, $\frac{d}{dx}[y]$ then:

- $[f(x)' \cdot g(x) \cdot h(x)] + [f(x) \cdot g(x)' \cdot h(x)] + [f(x) \cdot g(x) \cdot h(x)']$

15.1 Integrals

- Guide to definite integrals

16 Trig

- Summary of trigonometric identities
- Proof behind why $d/dx(\sin x) = \cos(x)$
- Chain rule on trig identity differentiation, for any equation of the form $(c \in R) : d/dx(\sin(cx))$
- More identities (overlap)
- Common Derivatives

Quick Cheat Sheet for me:

Is $\frac{\tan^3 x}{\tan x} = \tan^2(x)$ a valid operation?

True

Find $\frac{\sec(x)}{\tan(x)}$

$$\frac{\sec(x)}{\tan(x)} = \frac{\frac{1}{\cos(x)}}{\frac{\sin(x)}{\cos(x)}} = \frac{1 \times \cos(x)}{\cos(x) \times \sin(x)} = \frac{1}{\sin(x)} = \csc(x)$$